

Chapter

Chemical Bonding

1. In the molecule of water, oxygen atom has: MAIN 2024

- (a) One shared pair of electrons.
- (b) Three shared pair of electrons.
- (c) Two lone pairs of electrons.
- (d) One lone pair of electrons.

2. Ionic compounds in general possess both. SQP 2023

- (a) high solubility in polar and non-polar compounds
- (b) high melting point and low boiling points
- (c) directional bonds and low boiling points
- (d) high melting point and non-directional bonds

3. In the liquid state, X will : COMP 2022

- (a) become ionic
- (b) not conduct electricity
- (c) conduct electricity
- (d) be an electrolyte

4. A particular solution contains molecules and ions of the solute so it is a :

COMP 2023

- (a) strong acid
- (b) weak acid
- (c) Redox
- (d) salt solution

5. In the formation of magnesium chloride (by direct combination between magnesium and chlorine), name the substance that is oxidised and the substance that is reduced. COMP 2023

6. During electrolysis of acidulated water the gas liberated at the anode is (oxygen/hydrogen). MAIN 2024

7. What is the essential difference in the nature of compounds when two combining atoms. COMP 2022

(a) differ much in their electro-negativities ?

(b) don't differ much in their electro-negativities?

8. The electronic configuration of nitrogen is 2,5. How many electrons in the outer shell of a nitrogen atom are not involved in the formation of a nitrogen molecule ? SQP 2021

9. A bond formed between two atoms by sharing of a pair of electrons, with both electrons being provided by the same atom. COMP 2025

10. An ion formed by the loss of electrons from the neutral atom. MAIN 2021

11. An ion formed by the gain of electrons by the neutral atom. SQP 2023

12. Outermost shell of an atom. COMP 2021

13. Carbon tetrachloride is a (polar/nonpolar) covalent molecule. MAIN 2024

14. An element X is a metal with valency 1. Element Y is a non-metal with valency 1. SQP 2023

(i) Write equations to show how X and Y form ions.

(ii) If Y is a diatomic gas, write the equation for the direct combination of X and Y to form a compound.

15. Give reasons for the following: MAIN 2014

(i) Ionic compounds have high melting point.

(ii) Ionic compounds easily dissolve in water.

(iii) Inert gases do not form ions.

16. Match the Column A with Column B: MAIN 2024

	Column A		Column B
(a)	Water	1.	Lithium
(b)	Alkali metal	2.	Iodine
(c)	Halogen	3.	Covalent compound
(d)	Calcium oxide	4.	Acetic acid
(e)	Weak acid	5.	Ionic compound
		6.	Sulphuric acid

17. An atom P has 2,8,7 electrons in its orbits. It combines with Q having one electron in its valence orbit. MAIN 2023

- (a) What type of bond will be formed between P and Q ?
- (b) Write the formula of the compound formed.
- (c) Is it soluble in water ?

Solution

1. (c) Two lone pairs of electrons

- Oxygen has **6 valence electrons**.
- It forms **2 covalent bonds** with hydrogen, using 2 pairs of electrons.
- The remaining **4 electrons form 2 lone pairs**.

2. (d) high melting point and non-directional bonds

- Ionic bonds are **non-directional** (attraction occurs in all directions between ions).
- Strong electrostatic forces make **melting points high**.

3. (b) not conduct electricity

- Some compounds (like **covalent or molecular compounds**) do **not conduct electricity** in liquid form because they **do not produce free ions**.
- Only ionic compounds in molten state or in solution can conduct electricity.

4. (b) weak acid

- **Weak acids** are **partially dissociated** in water.
- They contain **both unionized molecules and ions**, making them **weak electrolytes**.

5. Magnesium is oxidised and chlorine is reduced.

6. Oxygen

7. (a) Atoms differ much in electronegativities:

- Form **ionic bonds** (complete transfer of electrons) or **polar covalent bonds** (partial transfer of electrons).

(b) Atoms don't differ much in electronegativities:

- Form **non-polar covalent bonds** (electrons are shared equally).

8. Two

9. Coordinate bond

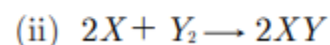
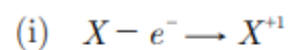
10. Cation

11. Anion

12. Valence shell

13. Non-polar

14.



15. (i) **High melting point:** Ionic compounds have **strong electrostatic forces** between oppositely charged ions, requiring a lot of energy to break.

(ii) **Solubility in water:** Water molecules **weaken the attraction** between ions, allowing the ionic compound to dissolve.

(iii) **Inert gases don't form ions:** They have **stable electronic configurations (full octet)**, so they neither lose nor gain electrons.

16.

	Column A		Column B
(a)	Water	3.	Covalent compound
(b)	Alkali metal	1.	Lithium
(c)	Halogen	2.	Iodine
(d)	Calcium oxide	5.	Ionic compound
(e)	Weak acid	4.	Acetic acid

17. (a) **Type of bond:** Electrovalent (Ionic) bond

(b) **Formula of the compound:** QP

(c) **Solubility in water:** Yes, it is soluble in water